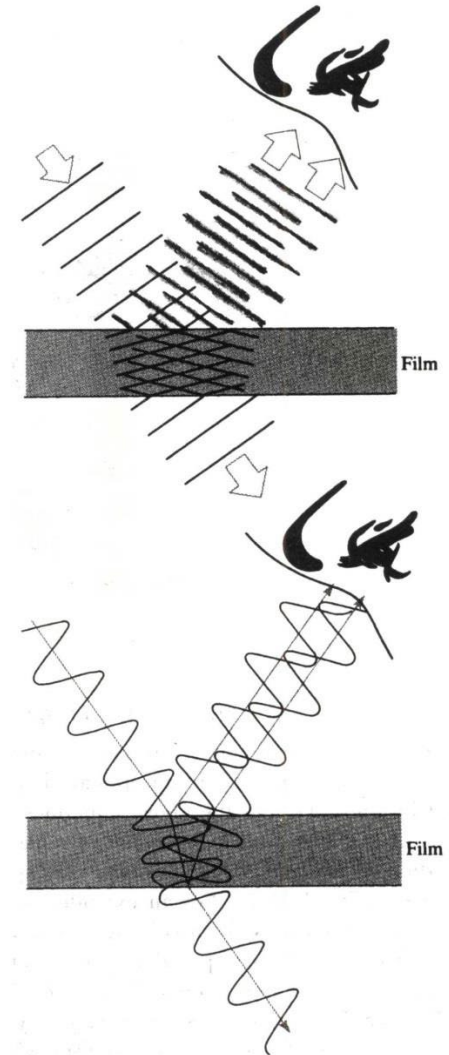


8.4 Amplitude-splitting interferometers

Suppose: a lightwave is incident on a half-silvered mirror or simply on a sheet of glass

- ☐ Part of the wave is transmitted and part reflected.
- ☐ Both the transmitted and reflected waves have lower amplitudes than the original one.
- ☐ The amplitude has been “split”.



Fringes of equal inclination:

Suppose:

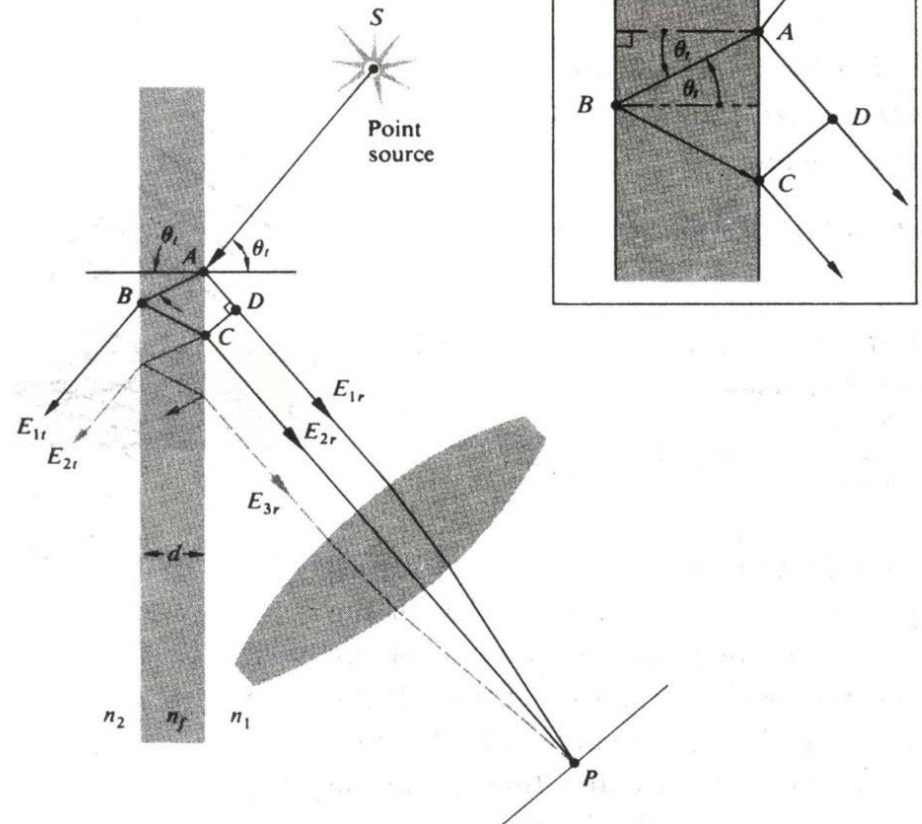
- ❑ the film is nonabsorbing;
- ❑ the amplitude-reflection coefficient at the interfaces are low enough, so only the first two reflected beams are considered.

Optical path-length difference:

$$\Lambda = n_f[(\overline{AB}) + \overline{BC}] - n_1 \overline{AD}$$

Solution:

$$\Lambda = 2n_f d \cos \theta_t$$



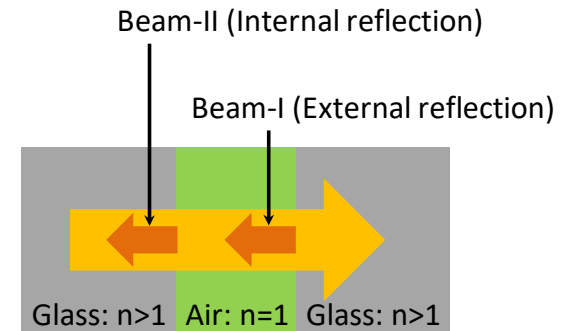
Case I: $n_1 = n_2 = n; n_f > n$ or $n_f < n$

A relative π phase shift

The phase difference:

$$\delta = k_0 \Lambda \pm \pi \xrightarrow{\text{for simplicity}} k_0 \Lambda - \pi = \frac{4\pi n_f d \cos \theta_t}{\lambda_0} - \pi$$

A relative π phase shift



When $\delta = 2m\pi$, an interference maxima:

$$d \cos \theta_t = \frac{\lambda_f}{4} (2m + 1)$$

When $\delta = (2m \pm 1)\pi$, an interference minima:

$$d \cos \theta_t = \frac{\lambda_f}{4} 2m$$

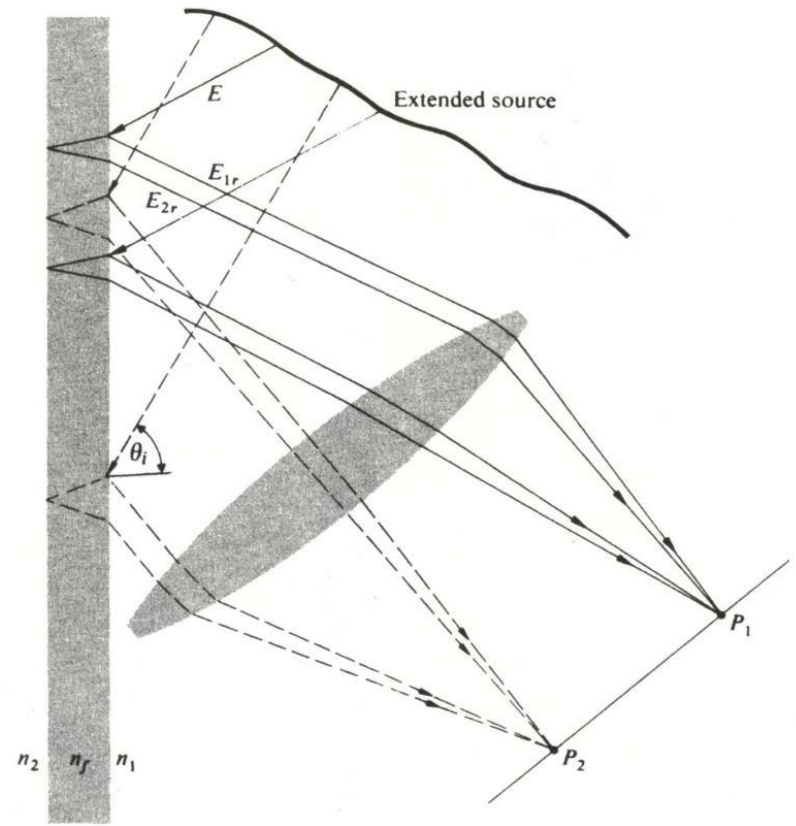
Case II: $n_1 > n_f > n_2$; or $n_1 < n_f < n_2$

no π phase difference.

Fringes of equal inclination

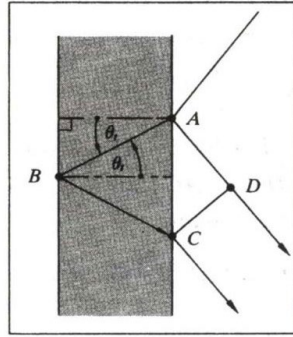
All rays inclined at the same angle arrive at the same point

- ❑ The angle θ_i or equivalently θ_t , determined by the position, will in turn control the phase δ .
- ❑ The fringes appearing at points P_1 and P_2 : fringes of equal inclination.
- ❑ Note: each source point on the extended source is incoherent.



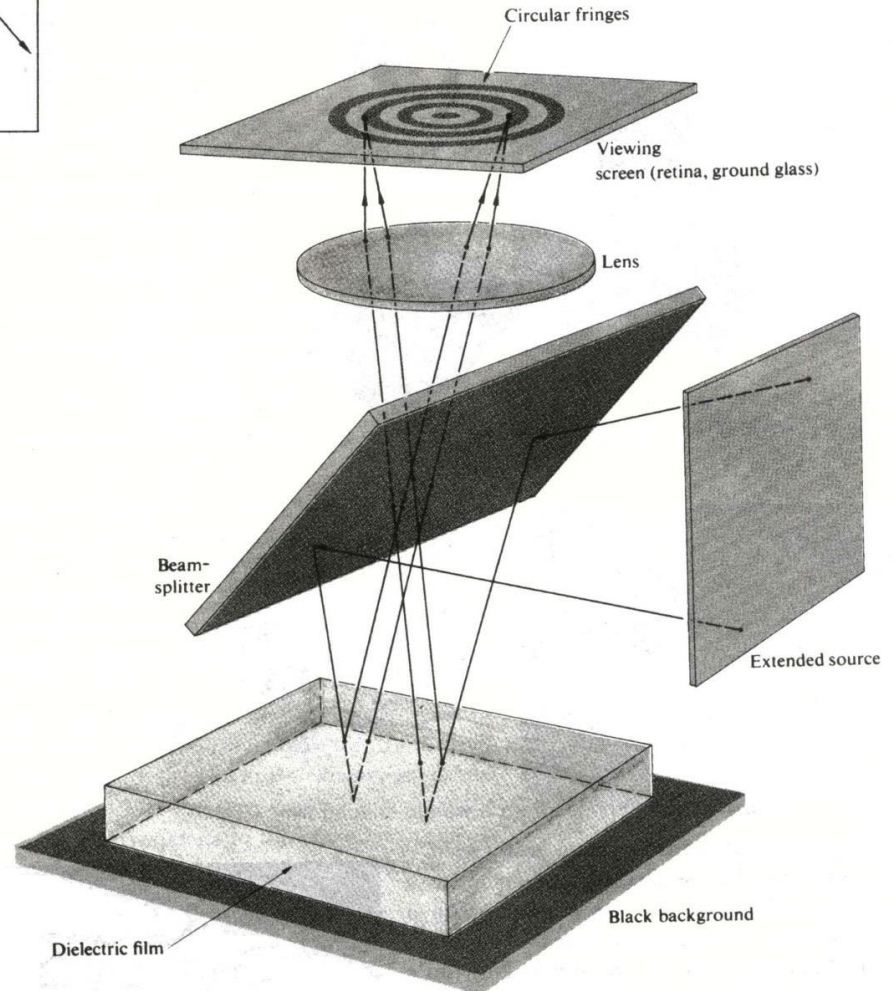
The separation $\overline{AC}$ between E_{1r} and E_{2r} :

$$\overline{AC} = 2d \tan \theta_t$$



As d increases, the separation of two rays increases. So a larger lens will be used to gather in both rays.

By reducing θ_t and therefore θ_i , the separation can be reduced, which is known as **Haidinger fringes**.



Homework

□ 9.1, 9.2, 9.4, 9.26, 9.29

Fringes of equal inclination

Optical path-length difference: $\Lambda = 2n_f d \cos \theta_t$

θ_i (θ_t) is dominant.

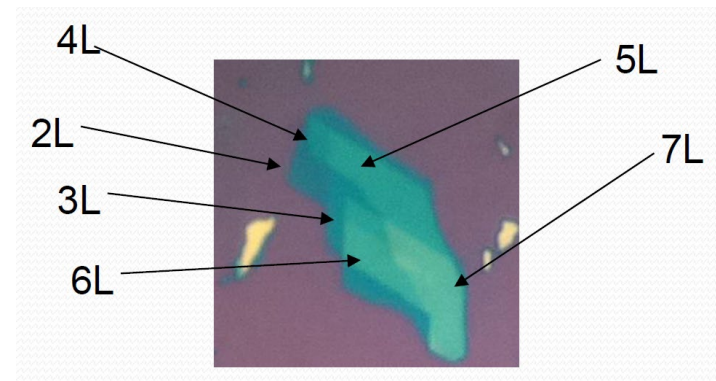
Fringes of equal thickness

Optical path-length difference: $n_f d$ is dominant parameter.

Examples:



Iridescence of soap bubbles



Layered MoS₂ materials

Fizeau fringes (斐佐干涉条纹):

Condition: nearly normal incidence; a thin wedge of small angle α ;

The optical path length difference:

$$\Lambda = 2n_f d \cos \theta_t$$

d : the thickness at a particular point ($d = x\alpha$)

In this case: $\theta_t \approx \theta_i \approx 0$, $\Lambda = 2n_f d$

An interference maximum (including a relative phase shift of π radians)

$$(m + 1/2)\lambda_0 = 2n_f d_m$$

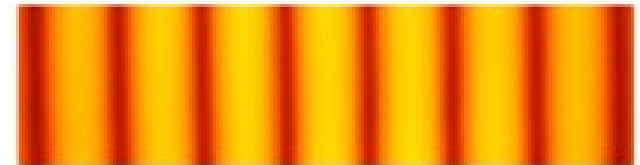
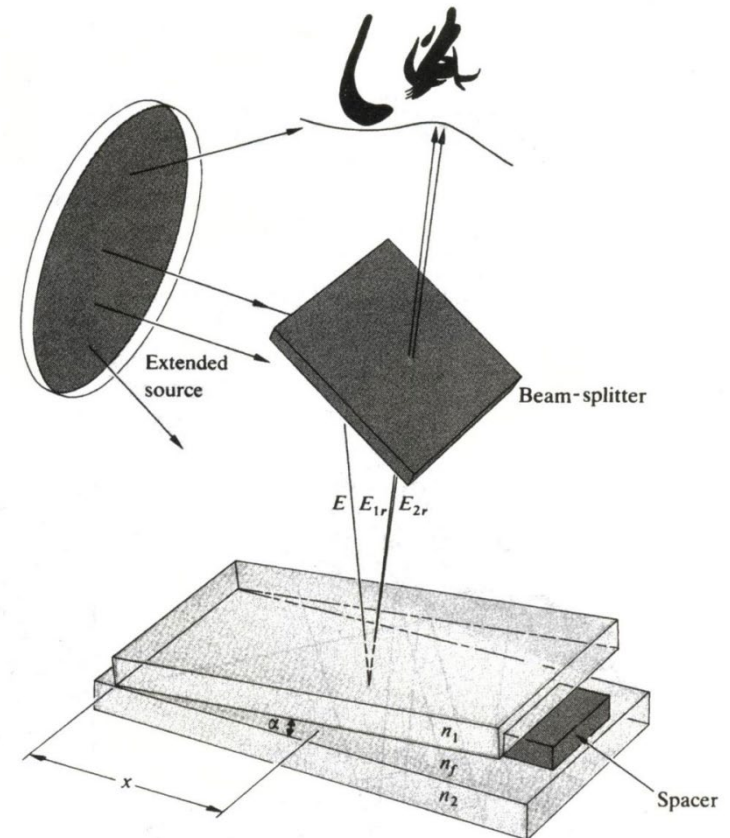
$$d_m = \left(m + \frac{1}{2}\right) \frac{\lambda_f}{2}$$

Maxima occurs at an odd multiple of a quarter wavelength.

$$(m + 1/2)\lambda_0 = 2n_f \alpha x_m$$

$$x_m = \left(\frac{m + 1/2}{2\alpha}\right) \lambda_f$$

Maxima occur at distances from the apex given by $\lambda_f/2\alpha$, $3\lambda_f/4\alpha$, and so on.



Monochromatic fringes

Example I: a wedge-shaped film made of liquid dishwashing soap

The black region at the top:

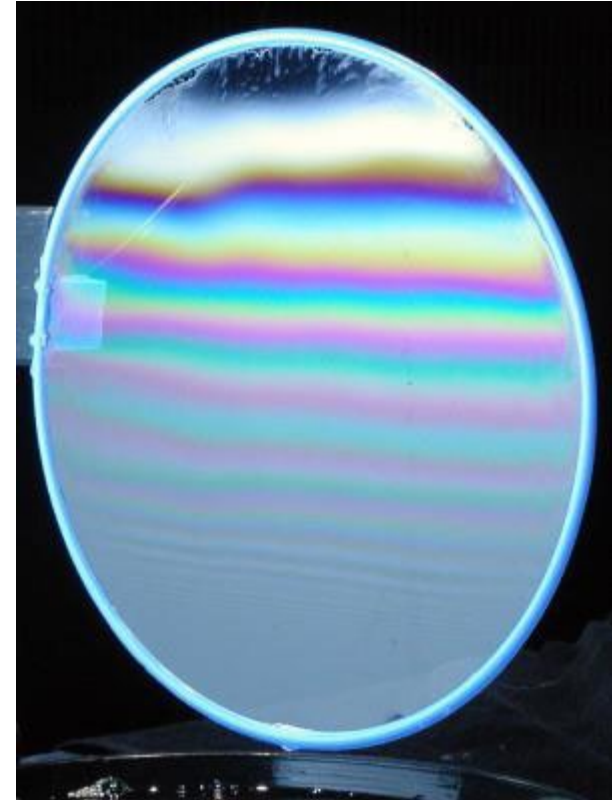
- ❑ The film thickness is less than $\lambda_f/4$; the reflected rays from two surfaces are $\lambda_f/2$ phase shift; the reflected rays are out of phase.

The soap film increases in thickness towards the bottom:

- ❑ Reflections from the front and rear of the soap film gradually undergo increasingly larger phase shifts.
- ❑ For example, when the thickness is $1/4$ wavelength, the two reflections is in phase and thereby giving a maximum reflected light intensity.
- ❑ As we continue to increase in thickness, every $1/4$ wavelength increase in thickness we move from a region of no reflection to a region of maximum reflection to no reflection again, etc.

The spacing of the light and dark reflection bands across the soap film:

- ❑ An indication of how the thickness varies.



Example II: Newton's ring

Condition: a lens is placed on an optical flat and illuminated at normal incidence with quasimonochromatic light.

The distance x :

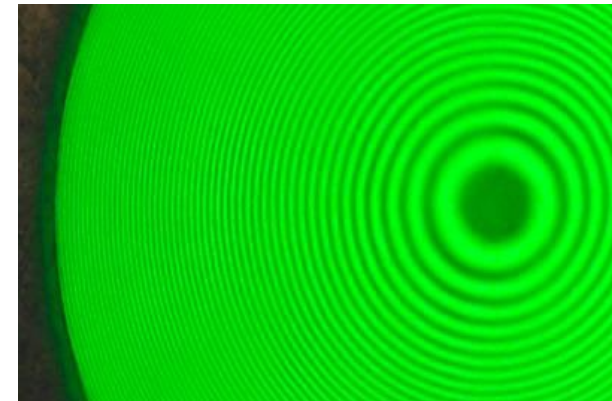
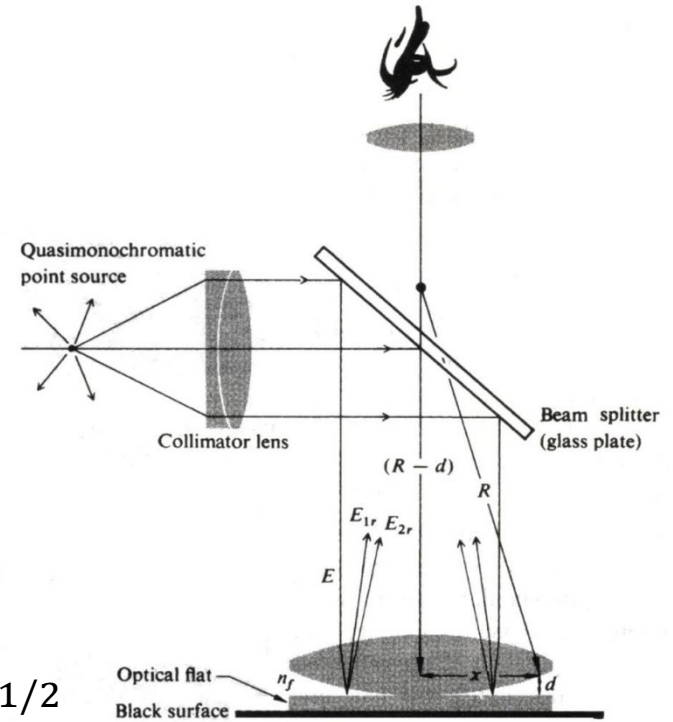
$$x^2 = R^2 - (R - d)^2 = 2Rd - d^2 \xrightarrow{R \gg d} 2Rd$$

The m th-order interference maximum occurs at:

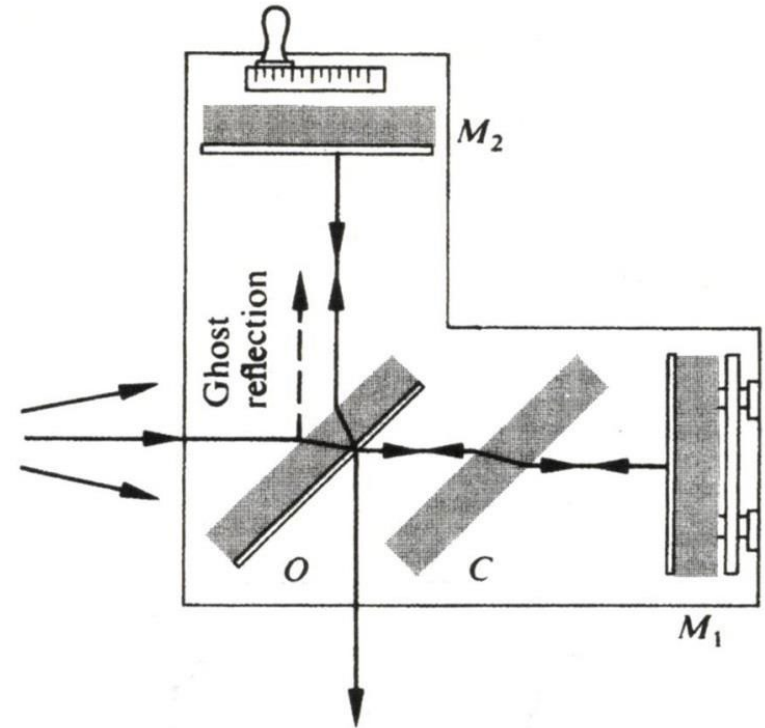
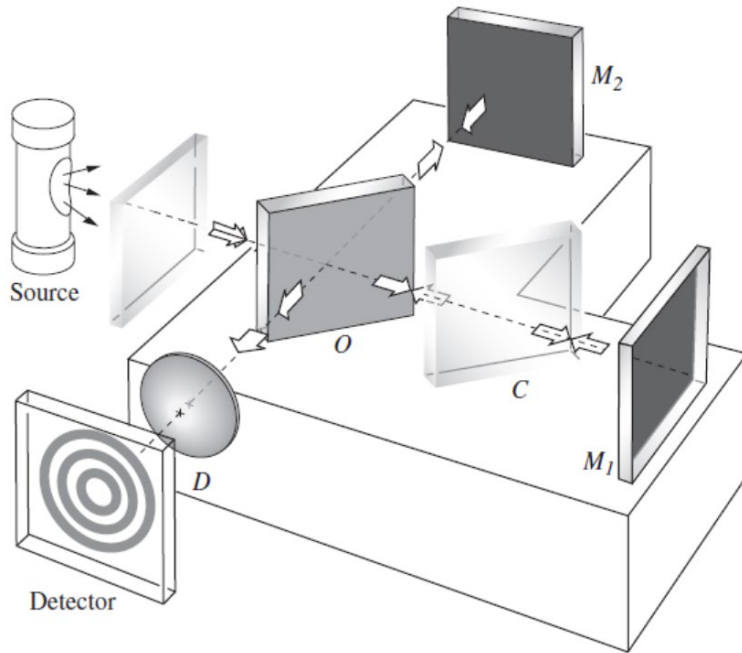
$$2n_f d_m = \left(m + \frac{1}{2}\right) \lambda_0 \Rightarrow x_m = \left[\left(m + \frac{1}{2}\right) \lambda_f R \right]^{1/2}$$

The m th-order interference minimum occurs at:

$$2n_f d_m = m \lambda_0 \Rightarrow x_m = (m \lambda_f R)^{1/2}$$



Michelson Interferometer



The compensator (C): an exact duplicate of the beamsplitter.

Function:

- ❑ Compensate the difference between any optical path and actual path.
- ❑ Negate the effect of dispersion, so a source with a very broad bandwidth will generate discernible fringes.

How Michelson interferometer works

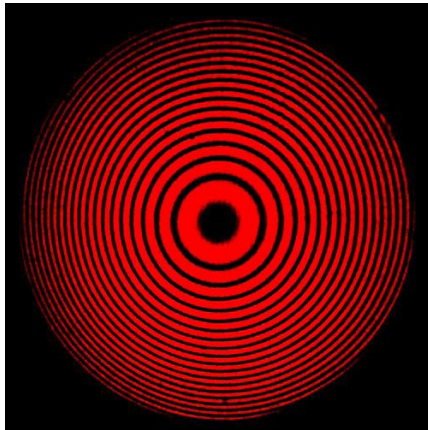
Fringes of equal inclination:

Destructive interference will exist (consider the relative phase shift of π radians) when

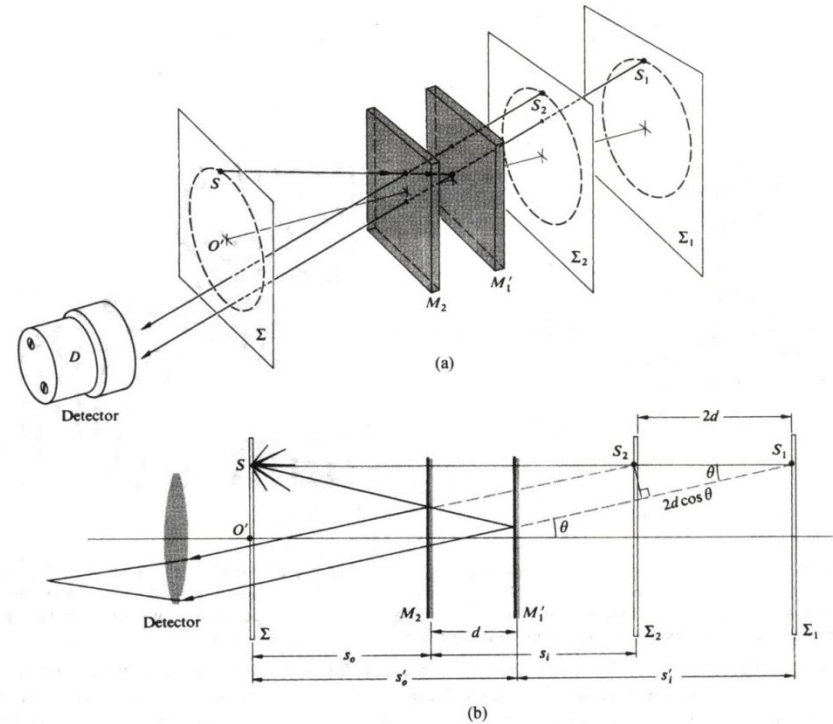
$$2d \cos \theta_m = m\lambda_0$$

The interference pattern:

A circular fringe system concentric with the central axis.



Note: $2d \cos \theta_m$ must be less than the coherence length of the source, so laser light will be easier than a tungsten bulb or a candle to use in demonstrating the interferometer.



A central dark fringe $\theta_m = 0$

$$2d = m_0 \lambda_0$$

Set $d = 10 \text{ cm}$; $\lambda_0 = 500 \text{ nm}$;

$$m_0 = \frac{2d}{\lambda_0} = 400000$$

Successive dark rings:

$$2d \cos \theta_1 = (m_0 - 1) \lambda_0$$

$$2d \cos \theta_2 = (m_0 - 2) \lambda_0$$

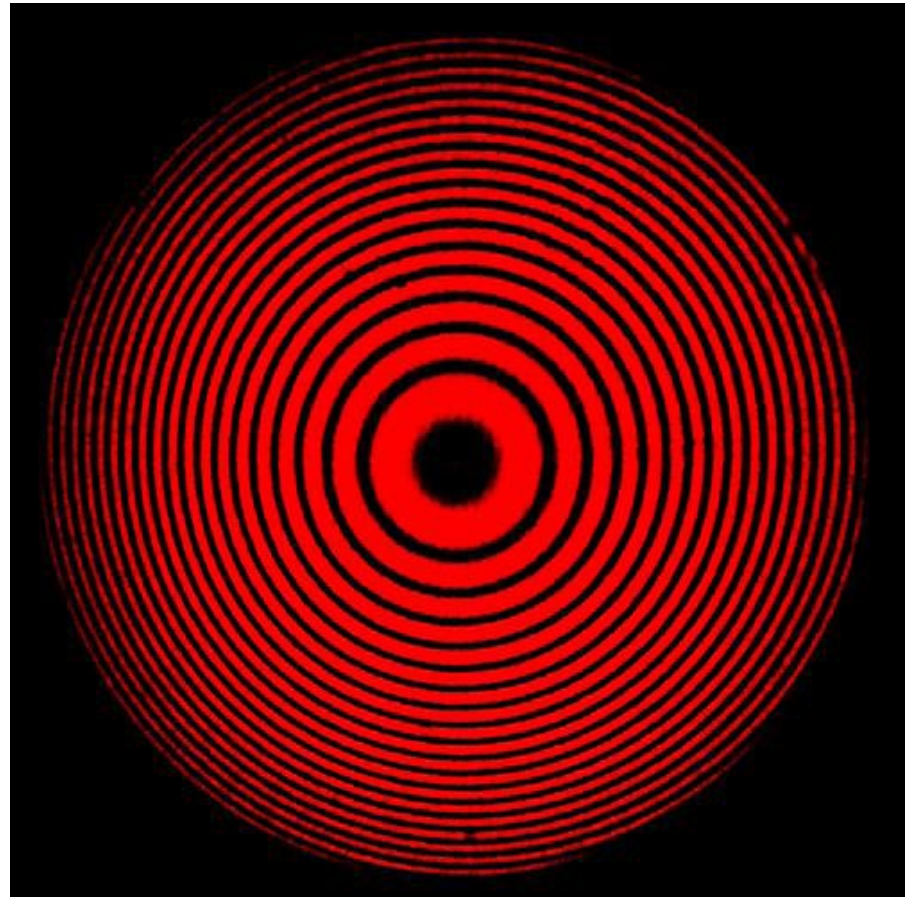
$\vdots$

$$2d \cos \theta_p = (m_0 - p) \lambda_0$$

For the p-th ring:

$$2d(1 - \cos \theta_p) = p \lambda_0$$

The sixth dark ring can be specified by stating that $p=6$, corresponding to $m=399994$ in the above case.



Wavelength measurement

As the movable mirror is displaced by $\lambda_0/2$, each fringe will move to the position previously occupied by an adjacent fringe.

$$\Delta d = N \left(\frac{\lambda_0}{2} \right)$$

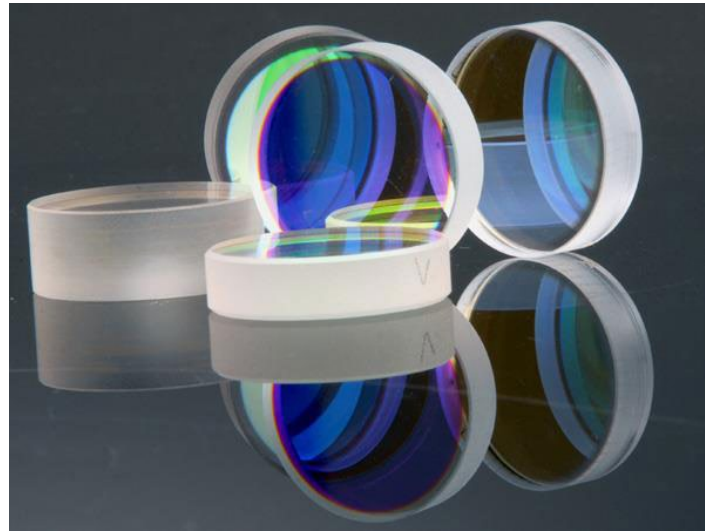
Nowadays this can be done fairly easily by electronic means:
spectrometer

Application of single and multilayer films

Coating to eliminate unwanted reflections off a diversity of surfaces, from showcase glass to high-quality camera lenses, are now commonplace.



Coated camera lenses



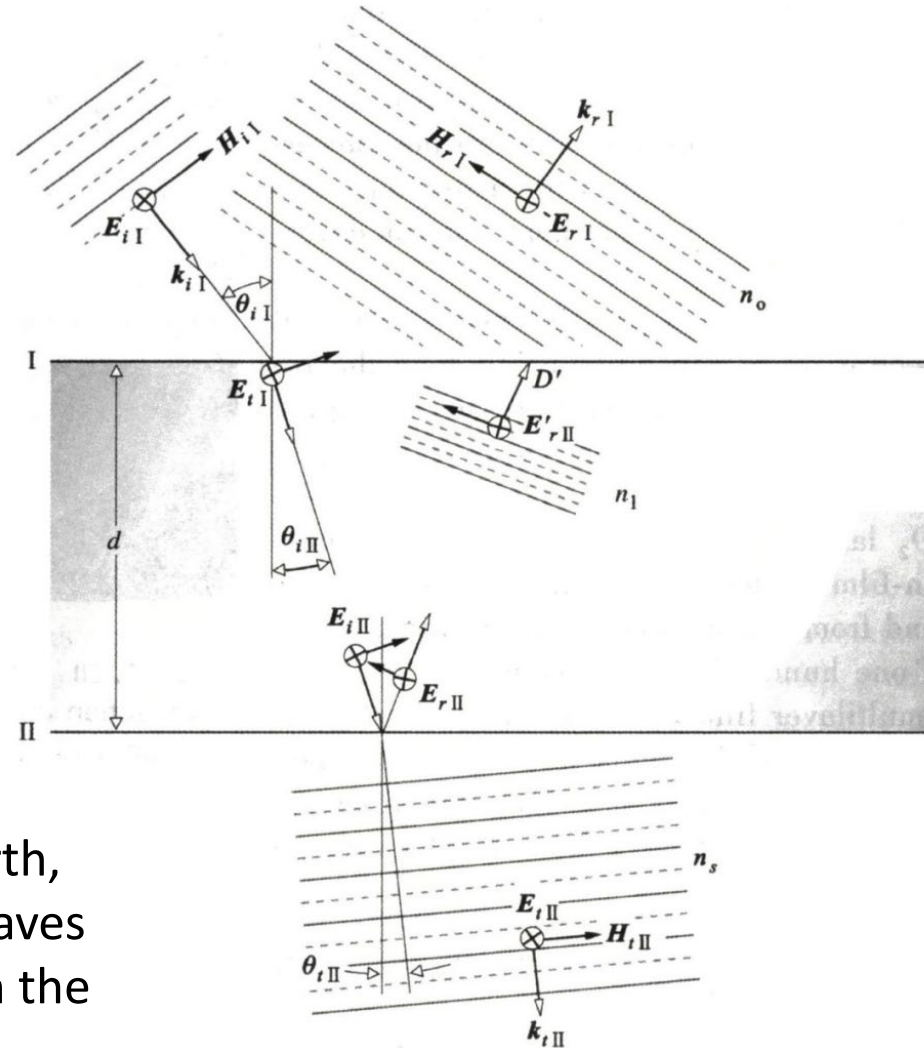
Dichroic mirrors

Mathematical Treatment

Consider the linearly polarized wave impinging on a thin dielectric film between two semi-infinite transparent media.

Boundary conditions: the tangential components of both the electric ($\vec{E}$) and magnetic field ($\vec{H} = \frac{\vec{B}}{\mu} = \sqrt{\frac{\epsilon_0}{\mu_0}} n \vec{k} \times \vec{E}$) be continuous across the boundaries.

Note: each wave E_{rI} , E_{rII}' , E_{tII} , and so forth, represents the resultant of all possible waves traveling in that direction, at that point in the medium.



At boundary I:

$$E_I = E_{iI} + E_{rI} = E_{tI} + E_{rII}'$$

$$H_I = \sqrt{\frac{\epsilon_0}{\mu_0}} (E_{iI} - E_{rI}) n_0 \cos \theta_{iI}$$

$$= \sqrt{\frac{\epsilon_0}{\mu_0}} (E_{tI} - E_{rII}') n_1 \cos \theta_{iII}$$

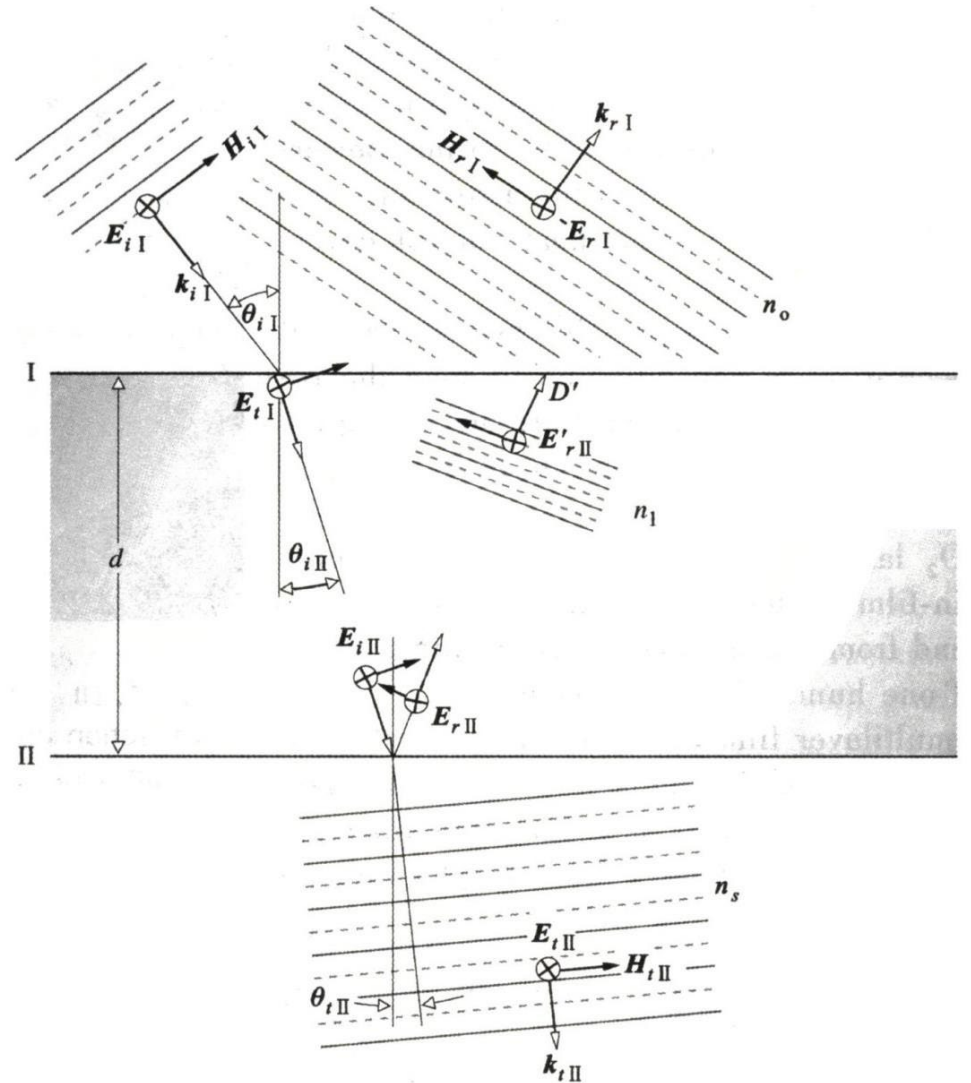
At boundary II:

$$E_{II} = E_{iII} + E_{rII} = E_{tII}$$

$$H_{II} = \sqrt{\frac{\epsilon_0}{\mu_0}} (E_{iII} - E_{rII}) n_1 \cos \theta_{iII}$$

$$= \sqrt{\frac{\epsilon_0}{\mu_0}} E_{tII} n_s \cos \theta_{tII}$$

$$\begin{bmatrix} E_I \\ H_I \end{bmatrix} = \begin{bmatrix} \cos k_0 h & \frac{i \sin k_0 h}{\gamma_1} \\ \gamma_1 i \sin k_0 h & \cos k_0 h \end{bmatrix} \begin{bmatrix} E_{II} \\ H_{II} \end{bmatrix}$$



$$k_0 h = k_0 (2n_1 d \cos \theta_{iII}) / 2$$

$$\gamma_1 = \sqrt{\frac{\epsilon_0}{\mu_0}} n_1 \cos \theta_{iII}$$

$$\begin{bmatrix} E_I \\ H_I \end{bmatrix} = \mathcal{M}_I \begin{bmatrix} E_{II} \\ H_{II} \end{bmatrix}$$

$$\mathcal{M}_I = \begin{bmatrix} \cos k_0 h & \frac{i \sin k_0 h}{\gamma_1} \\ \gamma_1 i \sin k_0 h & \cos k_0 h \end{bmatrix}$$

The characteristic matrix $\mathcal{M}_I$ relates the fields at the two adjacent boundaries.

For two overlaying films deposited on the substrate

$$\begin{bmatrix} E_{II} \\ H_{II} \end{bmatrix} = \mathcal{M}_{II} \begin{bmatrix} E_{III} \\ H_{III} \end{bmatrix}$$

For p layers, each with a particular value of n and h :

$$\begin{bmatrix} E_I \\ H_I \end{bmatrix} = \mathcal{M}_I \mathcal{M}_{II} \dots \mathcal{M}_p \begin{bmatrix} E_{(p+1)} \\ H_{(p+1)} \end{bmatrix}$$

$$\mathcal{M} = \mathcal{M}_I \mathcal{M}_{II} \dots \mathcal{M}_p = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}$$

At boundary I:

$$E_I = E_{iI} + E_{rI} = E_{tI} + E_{rII}'$$

$$H_I = \gamma_0(E_{iI} - E_{rI}) = \gamma_1(E_{tI} - E_{rII}')$$

At boundary II:

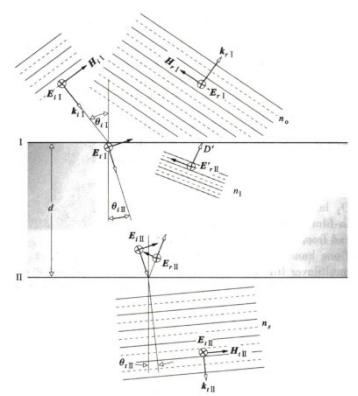
$$E_{II} = E_{iII} + E_{rII} = E_{tII}$$

$$H_{II} = \gamma_1(E_{iII} - E_{rII}) = \gamma_s E_{tII}$$

$$\gamma_0 \equiv \sqrt{\frac{\epsilon_0}{\mu_0}} n_0 \cos \theta_{iI}$$

$$\gamma_1 \equiv \sqrt{\frac{\epsilon_0}{\mu_0}} n_1 \cos \theta_{iII}$$

$$\gamma_s \equiv \sqrt{\frac{\epsilon_0}{\mu_0}} n_s \cos \theta_{tII}$$



$$\begin{bmatrix} E_I \\ H_I \end{bmatrix} = \mathcal{M}_I \begin{bmatrix} E_{II} \\ H_{II} \end{bmatrix} \quad \Rightarrow \quad \begin{bmatrix} (E_{iI} + E_{rI}) \\ (\gamma_0(E_{iI} - E_{rI})) \end{bmatrix} = \mathcal{M}_I \begin{bmatrix} E_{tII} \\ \gamma_s E_{tII} \end{bmatrix}$$

$$\Rightarrow \begin{cases} 1 + r = m_{11}t + m_{12}\gamma_s t \\ (1 - r)\gamma_0 = m_{21}t + m_{22}\gamma_s t \end{cases}$$

The amplitude coefficients of reflection and transmission

$$\Rightarrow \begin{cases} r = \frac{\gamma_0 m_{11} + \gamma_0 \gamma_s m_{12} - m_{21} - \gamma_s m_{22}}{\gamma_0 m_{11} + \gamma_0 \gamma_s m_{12} + m_{21} + \gamma_s m_{22}} \\ t = \frac{2\gamma_0}{\gamma_0 m_{11} + \gamma_0 \gamma_s m_{12} + m_{21} + \gamma_s m_{22}} \end{cases}$$

Antireflection (增透) coatings

Consider the important case of normal incidence: $\theta_{iI} = \theta_{iII} = \theta_{tII} = 0$

The reflection coefficient for a single film:

$$r_1 = \frac{n_1(n_0 - n_s) \cos k_0 h + i(n_0 n_s - n_1^2) \sin k_0 h}{n_1(n_0 + n_s) \cos k_0 h + i(n_0 n_s + n_1^2) \sin k_0 h}$$

$$R_1 = \frac{n_1^2 (n_0 - n_s)^2 \cos^2 k_0 h + (n_0 n_s - n_1^2)^2 \sin^2 k_0 h}{n_1^2 (n_0 + n_s)^2 \cos^2 k_0 h + (n_0 n_s + n_1^2)^2 \sin^2 k_0 h}$$

When $k_0 h = \frac{\pi}{2}$ (equivalent to $d = \frac{\lambda_f}{4}$)

$$R_1 = \frac{(n_0 n_s - n_1^2)^2}{(n_0 n_s + n_1^2)^2} \quad \text{which equals to zero, when } n_0 n_s = n_1^2$$

Example:

Cryolite ($n=1.35$) or MgF_2 ($n=1.38$) on glass ($n=1.5$): the reflectance of glass will reduce from 4% to 1%.

A double-layer, **quarter-wavelength** antireflection coating

$$k_0 h = \frac{\pi}{2} \Rightarrow \mathcal{M} = \mathcal{M}_I \mathcal{M}_{II} = \begin{bmatrix} 0 & i/\gamma_1 \\ i\gamma_1 & 0 \end{bmatrix} \begin{bmatrix} 0 & i/\gamma_2 \\ i\gamma_2 & 0 \end{bmatrix}$$

At normal incidence:

$$\mathcal{M} = \begin{bmatrix} -n_2/n_1 & 0 \\ 0 & -n_1/n_2 \end{bmatrix}$$

Reflectance:

$$R_2 = \left[\frac{n_2^2 n_0 - n_s n_1^2}{n_2^2 n_0 + n_s n_1^2} \right]^2$$

For $R_2 = 0$ at a particular wavelength: $\left(\frac{n_2}{n_1} \right)^2 = \frac{n_s}{n_0}$

Since $n_2 > n_1$; it is common practice to designate a (glass)-(high index)-(low index)-(air) system as gHLA.

Example: ZrO_2 ($n=2.1$); TiO_2 ($n=2.4$); ZnS ($n=2.32$) are used for H-layer; MgF_2 ($n=1.38$); CeF_3 (Cerium Fluoride, $n=1.63$) are employed as L-layer.

High reflectance: if $n_2/n_1 \gg 1$, $R = 1$

To further enhance the reflection, one can use a periodic quarter-wave stack system $g(\text{HL})^m a$

$$\mathcal{M} = \mathcal{M}_L \mathcal{M}_H \mathcal{M}_L \mathcal{M}_H \dots \mathcal{M}_L \mathcal{M}_H$$

$$\mathcal{M}_L \mathcal{M}_H = \begin{bmatrix} 0 & i/\gamma_L \\ i\gamma_L & 0 \end{bmatrix} \begin{bmatrix} 0 & i/\gamma_H \\ i\gamma_H & 0 \end{bmatrix} = \begin{bmatrix} -n_H/n_L & 0 \\ 0 & -n_L/n_H \end{bmatrix}$$

$$\mathcal{M} = \begin{bmatrix} (-n_H/n_L)^m & 0 \\ 0 & (-n_L/n_H)^m \end{bmatrix}$$

$$R = \left[\frac{n_0(n_H)^{2m} - n_s(n_L)^{2m}}{n_0(n_H)^{2m} + n_s(n_L)^{2m}} \right]^2$$

When $n_H/n_L \gg 1$ and m is large the reflectance can be very high.

More improvement can be achieved by adding one more high-index layer to the air side of the periodic system to become $g(\text{HL})^m \text{Ha}$ system.

Homework

□ 9.31, 9.33, 9.45, 9.46